

The Physiology of Climbing

If you want to climb harder, you must train smarter!

The next five chapters will describe more than one hundred exercises that will help you build a stronger, more stable and powerful, and hopefully injury-proof physique. You'll also learn how to organize your daily workouts and schedule your long-term training to get optimal results for the time and energy you have to invest. While it may be tempting to skip over this chapter on the physiology of climbing and jump right into the more enticing exercise-packed, photo-laced chapters ahead, I urge you not to!

The exercise science and training principles presented in this chapter provide an important educational foundation from which you can build long-term success in the gym and on the rocks. In possessing a 200-level understanding of exercise physiology—how muscles adapt to various training stimuli, what causes fatigue, how you can best train for strength, power, and endurance—you can go about executing an evidence-based workout and avoid getting lured into someone else's bad (and perhaps dangerous) training practices.

Here are several more reasons to read this chapter. You will learn:

- How muscle fibers are recruited in action, and how you can train for higher recruitment and more strength.
- How muscles adapt to training in order to give you more strength, power, and endurance.
- How energy is created in the muscles, and how you can train to increase power output and exercise capacity.
- What causes fatigue, and how you can train to develop a higher resistance to fatigue.
- The nine principles that build effective training and exercise programming.
- How training power, strength, and endurance are different disciplines yet intimately connected. You will learn how to improve in all three areas!
- How to recognize bogus “bro science” at the gym, call “b.s.” on invalid training spray, and screen online training advice to identify the nuggets of good information amid a sea of questionable advice.

Before getting started, I want to encourage you to delve deeper into this subject, if you find that acquiring this knowledge yields a unique sense of empowerment in training. Exercise physiology is an extraordinarily interesting field, and climbing science in particular is a rapidly growing topic of research. While this chapter is steeped in sports science, no field is ever a “settled science”—new research is born each month, better training practices are constantly in development, and ever deeper and deeper levels of understanding and nuance are awaiting the curious. Read, assimilate, and apply the information in this chapter (and the entire book!), and explore the footnoted research papers and some of the books on the Suggested Reading list at the end of this book. And be sure to bookmark my website, Training4Climbing.com—visit it regularly for new articles, research updates, and a dose of inspiration!

Muscle Function and Adaptations to Training

The basics of muscle function and adaptation to exercise is a good place to begin our study of exercise physiology. First, we'll examine the type of muscle contractions and the role muscles play in generating movement. Next, you'll learn the three types of muscle fibers and how fiber recruitment varies with the type of exercise stimulus. Lastly is the critical area of muscle adaptation to exercise, and how strength and endurance training protocols produce vastly different cellular response.

Types of Muscle Contraction and Muscle Roles

The production of movement involves three different muscular actions:

Concentric contraction: Muscle action in which the tension developed produces a shortening of the musculature, as in the biceps during the upward phase of a pull-up.

Eccentric contraction: Muscle action in which the muscle resists as it's forced to lengthen, as in the biceps during the lowering phase of a pull-up.

Isometric contraction: Muscle action resulting in no shortening of the muscle (no movement), as in the musculature of the forearm while steadily gripping a handhold.

Depending on the movement being performed, muscles take on one of the three following roles:

Agonist: The muscle or muscle groups causing an action to occur. For instance, the biceps and the latissimus dorsi muscle of the back are some of the prime movers in the pulling motions common to climbing. (Reference the muscle anatomy photos in appendix A to see the location of these muscle groups.)

Antagonist: The muscle or muscles providing an opposing force to the primary muscles in action. For example, in doing a pull-up, the triceps of the lateral upper arm oppose the action of the pulling biceps.

Stabilizer: The muscle or muscle groups that help stabilize the skeletal structures so that tension of the agonist (prime movers) can produce smooth, effective movement. In climbing, some of the most important stabilizer muscles are those supporting the scapula (e.g., serratus anterior and trapezius), and the finger extensor muscles, which stabilize the wrist. Also essential to athletic movement and postural control are the many muscles of the "core" that stabilize the spine.

Muscle Fiber Types

There are two distinct types of muscle fiber: slow twitch (ST) and fast twitch (FT). Fast-twitch fibers are further subdivided into two main subcategories, Type IIa and Type IIb, as shown in table 5.1.

Slow twitch (Type I): Slow-twitch fibers make up approximately 50 percent of the total skeletal muscle, though genetic variation can range from about 20 percent to 80 percent (Bloomfield, et al., 1992). These fatigue-resistant fibers generate energy via the aerobic pathway, consuming muscle glycogen, triglycerides (fat), and lactate to generate ATP (adenosine triphosphate) in the presence of oxygen. Slow-twitch fibers have high capillary and mitochondrial density, and high myoglobin and aerobic enzyme content. Energy production and fiber recruitment are relatively slow (compared to Type II fibers), however, such that slow-twitch fibers are predominantly used during relatively low-intensity activity such as easy to moderate climbing.

Fast twitch A (Type IIa): These fibers, also called fast oxidative fibers, contract quickly and intensely, and they are relatively resistant to fatigue. Type IIa fibers are red, due to the high capillary density and myoglobin content. These fibers possess high mitochondrial density and therefore can generate ATP quickly via oxidative metabolic processes. The primary fuel source is intracellular glycogen and glucose; however, small amounts of creatine phosphate and lactate may contribute to ATP production. These fibers excel at relatively high-intensity activity lasting only a few minutes.

Table 5.1 Muscle Fiber Type and Characteristics

Characteristics	Fiber Type		
	Type I (ST)	Type IIa (FTa)	Type IIb (FTb)
Contraction Time	Slow	Fast	Very Fast
Resistance to Fatigue	High	Intermediate	Low
Functional Duration	Long	Short	Very Short
Force Production	Low	High	Very High
Glycogen Content	Low	High	High
Mitochondrial Density	High	High	Low
Capillary Density	High	Intermediate	Low
Oxidative Capacity	High	High	Low
Glycolytic Capacity	Low	High	High
Fuel Sources	Triglycerides, Lactate	Glycogen, CP, Lactate	CP, Glycogen

Fast twitch B (Type IIb): These fibers, often called fast glycolytic fibers, have the fastest contraction time and generate energy almost entirely through the anaerobic system. These white-colored fibers have low myoglobin and capillary density, and few mitochondria. Anaerobic metabolic processes prevail, converting creatine phosphate and glycogen into ATP. Resultant metabolic by-products include lactate and—more problematic—hydrogen ions (H⁺), inorganic phosphate, and ammonia, among other things, all of which contribute to a rapid onset of fatigue. Type IIb fibers are recruited during brief, maximum movements such as gripping a tiny handhold, locking off on one arm, and performing a powerful lunge or campus move.

Your relative percentage of FT and ST muscle fibers is genetically determined and varies little in response to training. Naturally strong boulderers are likely gifted with a higher-than-normal percentage of FT fibers (among other things), whereas gifted endurance climbers likely have a higher percentage of ST fibers. Fortunately, ST fibers can be taught to act like FT fibers (and vice versa) through use

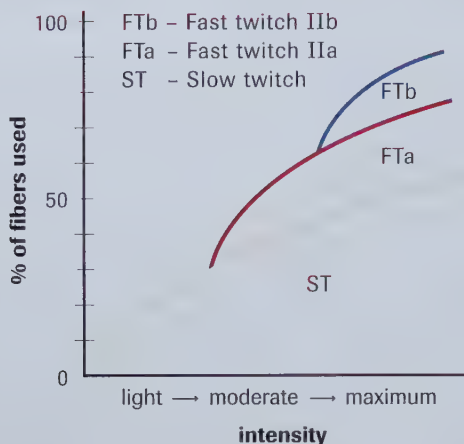
of certain training protocols (Chu 1996), and the recruitment and exercise capacity of each fiber type can be vastly improved via training. More on all these topics is upcoming!

Muscle Fiber Recruitment

Muscle fibers of the same type are organized into motor units. ST motor units innervate between 10 and 180 fibers, while FT motor units innervate up to 800 fibers (Bloomfield, et al. 1994). When a muscular contraction is triggered, motor units are recruited on an as-needed basis beginning with the smaller ST motor units. As muscular tension increases, a greater number of ST motor units will join in, and if the tension grows further, the stronger FTa motor units will begin to fire. Maximum muscular force is eventually achieved when a majority of motor units (ST, FTa, and FTb) are recruited into action (see figure 5.1).

This recruitment process explains the importance of strength training with high intensity and near-maximal weight (resistance)—it's only by recruiting the FTb fibers that you will markedly

Figure 5.1 Muscular Force Production



Maximum muscular force is achieved through an aggregate recruitment of ST, FTa, and FTb motor units.

increase strength and power. Thus, all the submaximal climbing in the world won't make you stronger; getting stronger at climbing requires near-maximal exercises such as hard bouldering, heavy weight lifting, hypergravity training, and such.

Later on in this chapter, you'll learn specific protocols (weights and reps) for developing maximum strength, power, and local strength-endurance, and in chapters 8 and 9 you'll learn the must-do exercises for climbers.

Muscular Adaptations to Strength Training

Two primary types of adaptations occur in response to strength training: neurological and morphological. Neural adaptations produce the initial strength gains resulting from use of a novel exercise or heighten exercises stimulus (see figure 5.2), so let's begin here.

NEUROLOGICAL ADAPTATIONS

The nervous system adapts to strength training in three ways: motor learning, motor unit synchronization, and disinhibition.

Motor learning: This first neural adaptation should sound familiar after reading chapter 4. During the initial work at a new exercise (say, uneven-grip pull-ups or even campus training), your primary limitation will be a lack of coordination and feel for the exercise. The first few workouts will yield rapid improvement as a result of motor learning and improved coordination among the prime mover, stabilizer, and antagonist muscles. Beyond this point, further strength gains will depend on other adaptations taking place.

Motor unit synchronization: Motor unit synchronization is the second neural adaptation that increases strength. Suppose you have acquired the coordination and motor skills needed for performing a given exercise—or perhaps you're adding a new exercise that requires no learning (such as hanging on a fingerboard). Initial training triggers motor units to fire in a rather random, asynchronous manner. Continued training, however, enhances motor unit synchronization; eventually most of the motor units will fire in unison, resulting in more strength and power.

Disinhibition: The final neural adaptation, disinhibition, is most important (and exciting) for intermediate to advanced climbers in search of gains in maximum strength and power. The neuromuscular system has a built-in feedback mechanism that acts as a safeguard during times of increasing force production. The Golgi tendon organ, located in the musculotendinous junction, is sensitive to the level of tension in the muscle, and in situations of high force it sends inhibitory signals that prevent further motor units from firing. In most individuals this protective response limits force production to some amount far below your absolute maximum-force-producing potential. It's like putting a restrictor plate on the engine of a race car to limit its top speed to 150 miles per hour, even though it's capable of 225. Fortunately, regular high-intensity training reduces the sensitivity of the Golgi tendon organ (disinhibition) and thus opens up a new level of maximum strength.

The difference between your maximum voluntary force and the absolute maximum capacity is called the strength deficit. Research has shown that significant gains in strength are possible by training to reduce these neural inhibitions. One study (Tidow 1990) showed that untrained individuals possessed strength deficits of up to 45 percent—that is, neural inhibition was reducing their maximum strength to almost half their absolute capacity. The study also revealed that targeted training by elite athletes reduced strength deficits to only 5 percent. Therefore, large gains in strength are possible without ever growing a larger, heavier muscle!

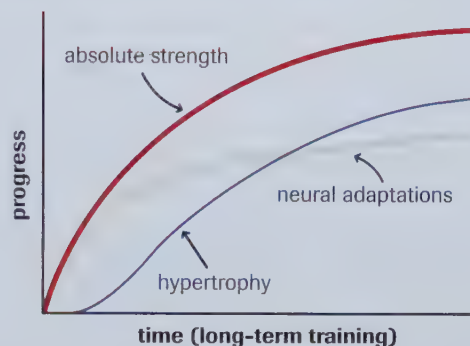
As a final note, the best type of training to produce disinhibition depends on the magnitude of your strength deficit. Intermediate climbers, who likely have larger strength deficits, would benefit most by training with heavy loads (weighted pull-downs and fingerboard hangs, hypergravity bouldering, HIT system, and so forth). Elite climbers with smaller strength deficits might realize further improvements only through a combination of high-resistance (hypergravity) and high-speed (reactive/plyometric) training.

MORPHOLOGICAL ADAPTATIONS

The primary morphological adaptations involve an increase in the cross-sectional area of the whole muscle and individual muscle fibers, a process called hypertrophy. As shown in figure 5.2, muscular hypertrophy contributes toward long-term gains in maximum strength.

Certainly, large muscles in the wrong place (e.g., legs, hips, chest, and shoulders) are a liability in a strength-to-mass ratio activity such as climbing. Even overdevelopment of the all-important pull muscles can be a bad thing if it's the result of exercising in a nonspecific way, such as a bodybuilding or CrossFit training program. For example, overdeveloped baseball-size biceps (a result of doing bicep curls with a heavy barbell) will have a negative biomechanical effect by preventing a tight lock-off when pulling a handhold in toward your shoulder.

Figure 5.2. Muscular Adaptations to Strength



Initial gains in strength result mostly from neural adaptations, whereas long-term gains depend to some degree on hypertrophy.

Still, any muscular hypertrophy occurring in the forearms, arms, and back resulting from climbing-specific training should be viewed as a good thing. In fact, an experienced climber who has been training for a long time and doesn't realize a little hypertrophy probably isn't training effectively or eating right.

It's interesting to note that a highly trained neuromuscular system is not absolutely necessary for being a strong climber. As mentioned in chapter 1, a small number of individuals possess tendon insertion points at a larger distance from the joint (axis of rotation) than the rest of us with average genetics. These gifted people will exhibit what seems to be amazing strength given their modest body builds. Other genetic factors, such as having a slight build, a large positive ape index, or an unusually high percentage of fast-twitch fibers, may further enhance their physical prowess. With this in mind you can see why these rare climbers are incredibly strong regardless of the type of training, if any, they engage in.

Muscular Adaptations to Local Endurance Training

Muscular adaptations that result from endurance-oriented training are fundamentally different from

the strength-training adaptations (described above), and are mainly confined to the FTa and ST fibers. Engaging in regular, rigorous local endurance training—“local” meaning in a specific muscle group, like the forearm flexors—produces a variety of adaptations both in the short and long term. Engaging in relatively high-intensity exercises such as Frenchies, bouldering 4x4s, hangboard repeaters, and climbing intervals will yield positive adaptations that are revealed in as little as a couple of weeks (buffering and enzymatic changes) up to many months and years (via angiogenesis). Detailed below are a few of the most significant adaptations to endurance exercise; however, there are several other more subtle cellular alterations that also contribute.

Intracellular buffering, oxidative enzymes, and glycogen supercompensation: Beginning a mesocycle of local endurance training (e.g., forearm strength-endurance and pull-muscle power-endurance) will yield noticeable improvement in only a few sessions. This initial windfall of endurance gains comes mostly by way of improved intracellular buffering of hydrogen ions (H⁺), development of beneficial glycolytic and oxidative enzymes, and glycogen supercompensation. Increases in these areas largely cease after a few weeks, so further gains in local endurance depend on other adaptations (below).

Mitochondria biogenesis: Increased mitochondria density—in both the ST and FTa fibers—is the most fundamental adaptation induced by endurance training. Mitochondria are the intracellular energy factories that convert glucose, glycogen, fatty acids, and lactate into ATP. Four weeks of endurance-oriented training (or climbing) can increase mitochondria density by up to 50 percent. Although small increases in mitochondria efficiency can continue long term, actual gains in mitochondria content approach a steady state after four to five weeks of training (Terjung 1979).

Angiogenesis: Endurance exercise increases the number of capillaries surrounding individual fibers, a process called angiogenesis. Increased capillarity improves oxygen exchange between capillary and fiber, thus enhancing the rate of energy production in ST and FTa fibers. As a relatively slow biological process, angiogenesis provides a pathway to longer-term gains in endurance.

Energy Systems

One of the many major additions to this edition of *Training for Climbing* is the concept of energy system training—that is, designing exercise protocols and workouts to target a specific energy system. This is

Table 5.2 Physiological Adaptations to Training

	Maximum Strength Training	Local Endurance Training
Muscle Fiber Size	No change or increase	No change or decrease
Fiber Recruitment	Increase	No change
Mitochondrial Density	No change	Increase
Capillary Density	No change	Increase
Cellular ATP-CP	Increase	No change
Cellular Glycogen	No change	Increase
Intracellular Buffering	No change	Increase

Adaptations from maximum strength training versus local endurance training.

Table 5.3 Energy Systems

Energy System	Energy Source	Duration of Steady-State Exercise	Power Output
Anaerobic Alactic	ATP-CP	1–12 seconds	Very high
Anaerobic Lactic	Muscle Glycogen	12 seconds–about 2 minutes	High
Aerobic	Fatty Acids, Glycogen, Lactate	2 minutes–about 2 hours	Low to Moderate

a powerful distinction that you can leverage to train far more effectively, given a sufficient understanding of the bioenergetics involved. If this sounds exciting, read on!

Energy systems refer to the metabolic pathways that produce ATP for muscular contractions. There are three major energy pathways, and all three contribute to ATP production at any given moment. Depending on the energy needs of exercise, however, just one or two energy pathway(s) dominate (see table 5.3). Let's take a closer look at each energy pathway and how they contribute toward power output in exercise and climbing.

ATP-CP (Anaerobic Alactic)

The ATP-CP system, also known as the anaerobic alactic energy pathway, provides rapid energy for brief, intense movements such as a few hard pull-ups, a powerful lunge, or a maximal contraction of the finger flexors. In maximal exercise the stored supply of ATP and CP (creatine phosphate) will diminish in just ten to twelve seconds, resulting in a drop in power output as the glycolytic (anaerobic lactic) system takes over energy production. This explains why it's virtually impossible to perform your maximum campus training movements for more than about ten seconds.

A strong anaerobic alactic energy system is important in climbing, because it's called upon to produce the maximum power output needed for a

crux move or a lunge or to sustain a maximal finger grip for a few seconds. Fortunately, depleted creatine phosphate stores can recover quickly during submaximal climbing and brief rest periods. What's interesting—and relatively unknown among coaches and athletes—is that creatine phosphate resynthesis is an aerobic process. Therefore, a strong aerobic energy system will support a high rate of creatine phosphate resynthesis—important for climbing intermittent near-maximal moves separated by only a few easier moves or a brief marginal rest.

The bottom line: Your level of aerobic conditioning plays an important role in your ability to do repeated bouts of anaerobic alactic-fueled exercise or maximal moves. Recent research has confirmed this concept in showing that aerobically trained climbers (who engaged in regular nonspecific aerobic activity such as running) recovered faster than climbers who did no generalized cardiovascular training (Schöffl 2006). Furthermore, Fryer (2015) has shown that the finger flexor muscles of elite climbers use (and replenish) oxygen at a higher rate than non-elites.

Glycolytic (Anaerobic Lactic)

Sustained high-intensity exercise (or climbing) lasting between about twelve seconds and two minutes is principally powered by the anaerobic lactic energy pathway. The process of fast glycolysis can rapidly regenerate ATP in an oxygen-free

(anaerobic) environment, which is the common state of the forearm flexor muscles during high-intensity isometric contractions (as in sustained gripping of difficult holds) and the large muscles of the arms and back when pulling many successive hard moves without rest. Muscle contractions of as little as 15 percent of maximum voluntary contraction (MVC) begin to impede blood flow, and contractions of more than 50 percent of MVC completely occlude blood flow (Köstermeyer 2000), so it's easy to understand why the anaerobic lactic energy system is so often called upon in climbing.

Common hallmarks of a hardworking anaerobic lactic energy system are muscular pain, the “pump,” a growing shortness of breath, and increasing fatigue (and drop in power output) between thirty seconds and two minutes. At about sixty seconds into sustained near-maximal climbing, the anaerobic lactic system begins to fade quickly, and by two minutes the aerobic energy pathway becomes the dominant source of ATP, thus compelling a sharp drop in power output.

The choking off of the anaerobic energy pathway between one and two minutes of sustained, difficult exercise is due to the metabolic by-products of the anaerobic splitting of glucose to create ATP. Although a complex process (and still somewhat controversial), we know that the anaerobic metabolism of glucose generates a number of by-products including pyruvate and hydrogen ions (H^+), which are generated at a rapid rate, along with lesser amounts of other metabolites such as inorganic phosphate, reactive oxygen species, and ammonia. While all these metabolites may contribute to fatigue, it's the rapidly increasing H^+ concentration that most threatens homeostasis and ultimately leads to cessation of this energy pathway.

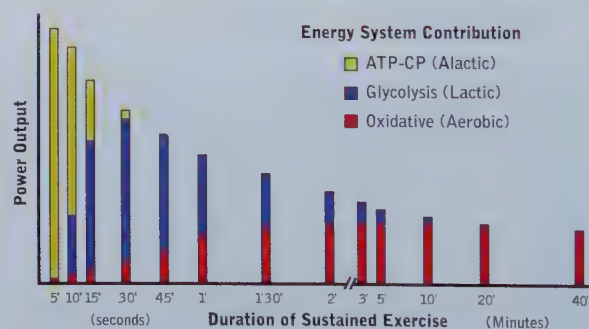
This increasing intracellular acidosis (due to H^+ production) is likely the cause of the muscle burn you feel; more critical, however, rising acidosis hampers the function of glycolytic enzymes, which results in a sharp drop in anaerobic energy production. Fortunately, there are intracellular buffers that

can consume H^+ (for a short time) and slow the rising tide of acidosis. Interestingly, the metabolite pyruvate is one such buffer, as one molecule of pyruvate can attract two H^+ ions to create lactate. Even more fascinating is the fact that lactate can pass into a nearby ST fiber and be used as fuel for aerobic metabolism. The bulk of the lactate, however, exits the cell into the blood and circulates to the liver to be converted back into glycogen, to the brain or other organs to support local energy production, or to other muscles where it can be used for aerobic energy production. This helps explain why aerobically trained climbers can recover more quickly between bouts of strenuous (anaerobic) climbing.

Aerobic

Sustained activity lasting longer than two minutes relies primarily on the aerobic (with oxygen) pathway, as energy produced via the anaerobic alactic and lactic systems quickly wanes. This transition to aerobic energy production is evident by a marked drop in power output due to the slower rate of ATP synthesis via the aerobic energy system (see figure

Figure 5.3 Muscular Energy Production



Brief, maximal muscular action is fueled by ATP-CP. After about twelve seconds, anaerobic lactic energy production becomes the primary energy source. Exercise lasting longer than two minutes depends mostly on aerobic energy production, although power output is only a fraction of maximal.

5.3). Fortunately, the only metabolic by-products of the aerobic system are CO₂, water, and heat—therefore, lower-intensity aerobic activity can continue for as long as oxygen and energy substrate are available (primarily muscle glycogen and fatty acids) and heat can be removed from the working muscle. Interestingly, the lactate produced by anaerobic glycolysis is another fuel source for aerobic energy production, and therefore lactate is a kind of molecular “holding tank” that bridges the anaerobic and aerobic energy systems.

Causes of Fatigue

After reading the previous sections on muscle fiber recruitment and the three energy systems, you should have a good understanding of some of the causes of the fatigue you experience in training and climbing. On a cellular level, depletion of ATP-CP, lack of oxygen (hypoxia), and increasing acidosis are primary causes of acute fatigue and diminished power output, while on a systemic basis, glycogen depletion and limitation of the circulatory system result in general fatigue. This textbook conceptual model of fatigue is called the Cardiovascular/Anaerobic Model. But here’s where the plot thickens . . .

While all the fatigue-inducing factors outlined above are certainly at play during intense exercise, the idea that fatigue results directly and absolutely from the muscles is now being challenged. New research suggests that it is actually the brain that initiates stress signals (the pain you feel) and triggers the “stopping thoughts” that arise in the conscious mind, based on cues received from the muscles. This new Central Governor Model (CGM) of fatigue was first proposed by Dr. Timothy Noakes, a marathon runner and a professor of exercise and sports science at the University of Cape Town, and it is now gaining traction with other researchers, coaches, and athletes. (I first wrote about the possible role of the central governor in my 2010 book, *Maximum Climbing*—pick up a copy for a study of brain training for climbing!)

According to the CGM, fatigue is a sensation sent by the unconscious mind to the conscious mind to prevent loss of homeostasis. (Homeostasis is the tendency of the body to seek and maintain a condition of balance or equilibrium within its internal environment; a simple example being the body’s ability to maintain an internal temperature around 98.6°F/37°C.) So what you feel during strenuous exercise—“my arm muscles hurt” or “my legs ache”—is actually the brain sending a message to slow down so as to avoid serious injury or death. Similarly, the brain involuntarily reduces neural drive (recruitment) in order to reduce power output and protect the body from irreversible damage.

The Central Governor Model remains somewhat controversial among sports scientists, but emerging research is lending support and increasing the number of advocates. As the theory goes, power output during exercise is regulated in the brain by a “central governor.” Efferent control of the muscles is based upon a black box calculation that integrates afferent feedback signals from the muscles and heart, as well as sensory feedback of the external environment (weather conditions, terrain, and such), knowledge of ongoing results (competitive place or proximity of the endpoint of performance), and unconscious calibration based on past experience. Integrating all this data, the central governor is constantly assessing how the organs and muscles are faring and what modulations are needed to maintain homeostasis. So it’s the brain that interprets increasing acidosis and hypoxia, elevated body temperature, and decreasing glycogen supply, among other signals, as being stressful to the body, and it’s the brain that in turn issues pain sensations and stopping thoughts.

You could argue that there’s little difference between the Cardiovascular/Anaerobic Model and the Central Governor Model, given that both models ultimately lead to fatigue based on what’s happening in the muscles. However, the CGM’s theory that the brain is the sole governor of fatigue is a crucial distinction with powerful implications.

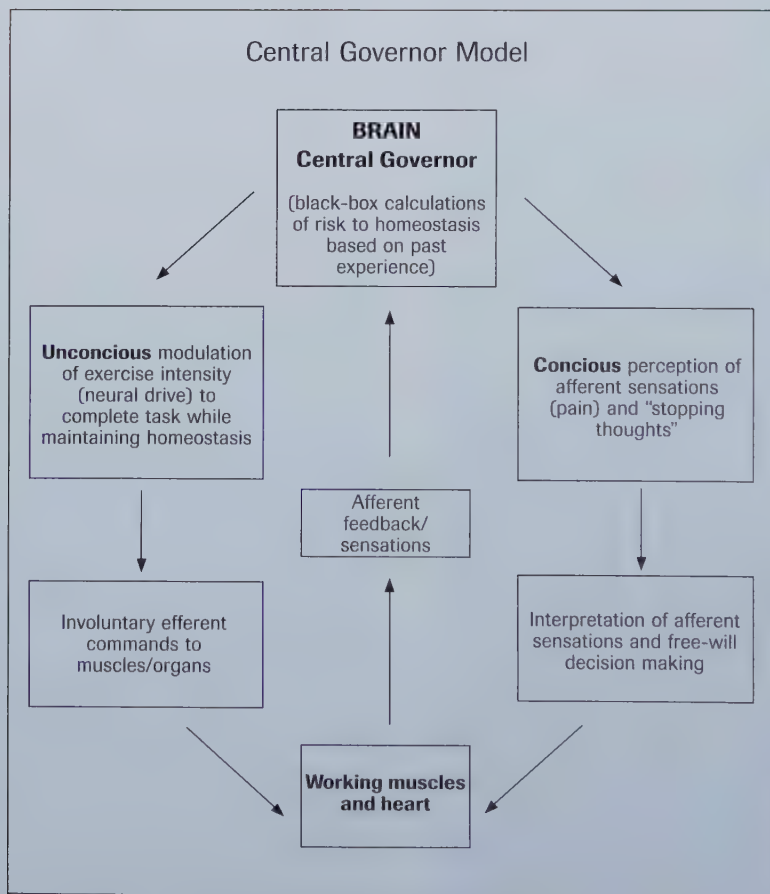
Since fatigue is not an absolute physiological event within the muscle, but actually a brain calculation and efferent response, it is possible to extend performance in the face of the brain's distress signals!

One recent study that supports this theory tested a group of cyclists performing a rigorous one-hour time trial. Australian researchers measured power output and muscle fiber activation during a mock time trial in which the cyclists performed six one-minute maximum sprints interspersed throughout the hour of sustained exercise. The results revealed that power output and muscle activation decreased with each successive sprint (as

you would expect). Surprisingly, however, the sixth sprint, performed during the last minute of the time trial, showed a significant increase in power output and muscle fiber recruitment! If the muscles themselves were the end-all in terms of fatigue signals (i.e., fatigue as an absolute result of high blood acidosis, dwindling ATP, increasing oxygen debt, and such), then the cyclists would have been physiologically unable to increase power output in the final stage of the time trial.

Per the CGM, here's what likely happened. Knowing the end of the time trial was near, the brain reassessed its estimates of energy reserves and

Figure 5.4 The Central Governor Model



afferent signals from the muscles and heart, and determined that it could allow a higher work output during the final minute of the time trial (without loss of homeostasis and risk of injury or death). Furthermore, the cyclists' knowledge that the final sprint would be followed by cessation of the grueling exercise may have empowered some of the cyclists to overcome, by the conscious power of will, the brain's pain signals and stopping thoughts. Applied to climbing, the Central Governor Model helps explain those instances when you climbed onward through massive fatigue to successfully clip the anchors on a sport climb or realized just enough of a "second wind" to complete a multi-pitch or big-wall ascent.

It's interesting to note that two parallel pathways operate within the Central Governor Model: the objective realm of the material world governed by the forces of mechanistic determinism, and the subjective realm governed by will (see figure 5.4). As exemplified in the preceding study (and perhaps in your own experience), willpower of the mind can exert a causal influence over determinism. While I believe that extreme physical activity is sometimes modulated solely by objective physiological constraints (per the Cardiovascular/Anaerobic Model), it is likely that performance most often falters because we obey or succumb to the brain's overly conservative signals to cease physical activity. The best climbers, then, are not necessarily the absolute strongest physically—instead they likely possess uncommon mental strength and willpower that enables them to hang on longer and continue upward, in spite of the brain's rising chorus of pain sensations.

In conclusion, we can distill two important training implications from the Central Governor Model. First and foremost, you can enhance and extend performance through the power of your free will. With consistent, disciplined effort to employ greater willpower in all you do, this largely untapped power of the mind will grow stronger and more capable of carrying you through extreme situations.

Secondly, engaging in high-intensity or long-duration training (and climbing) to a high level of fatigue will recalibrate the brain to be less conservative in directing work output during extreme physical activity, thus enabling you to perform longer and nearer your body's true absolute limit.

Let's take a brief look at how you might take this knowledge of the central governor and apply it to training for climbing.

Strength and Power Training to Recalibrate the Central Governor

Muscular strength and power are highly distinct functions that involve brief, intense force production. Absolute muscular strength, often called "limit strength," is the ability to exert maximum force in a single all-out effort (irrespective of time), as in crimping on a minuscule handhold or pulling hard through a strenuous move. Power is the application of force with velocity—think of it as explosive strength—as in making a fast, powerful reach upward or throwing a lunge.

Given the brief duration of maximum strength and power at work, the factors that lead to fatigue and limit performance are vastly different from those of muscular endurance. A primary limitation in maximum-strength and power output is the supply of ATP-CP stored in the muscle—the supply of these fuels limits maximal force production to less than twelve seconds (although ATP-CP is continually synthesized within the muscles and replenishes in just three to five minutes of rest).

While the central governor plays a lesser role in modulating such maximal movements, the brain is a major factor in directing application of strength and power. It's ultimately the brain that dictates recruitment of muscular motor units via neural drive (the amplitude of the impulses impelling recruitment) and by cycling motor units within a muscle in order to sustain steady force production for as long as possible. As an example of this amazing process, consider the act of holding steady a

one-arm lock-off position for a few seconds. During this maximal activity, the brain cycles the firing of motor units to maintain a steady lock-off; it rests some motor units (groups of muscle fibers) while it fires others. Maximum-strength and power production are greatly influenced by the quality of coordination and synchronization of this motor unit recruitment.

Two final factors that limit strength and power production are the Golgi tendon organ (GTO) and willpower. The GTO is part of the autonomic nervous system and protects tendons by limiting the recruitment of motor units below a certain threshold amount. In a manner quite similar to the central governor, the GTO restricts power output to a level far below the likely point of catastrophic tendon damage. Unlike the central governor, however, GTO function is completely involuntary and essentially instantaneous in reaction since its neural pathway only travels from the sensory receptors in the tendon to the spinal cord and back, and not the longer round-trip to the brain and back. Still, the sensitivity of the GTO can be influenced through training. Specifically, reactive training methods (more on this later) will somewhat disinhibit the GTO, to allow a higher level of maximum strength and power output.

Muscular Endurance and Stamina to Recalibrate the Central Governor

Let's first define muscular endurance and stamina, since some people incorrectly use these terms interchangeably.

Stamina is the ability to engage in low- to moderate-intensity activity for an extended period, such as running for an hour or two, or even climbing all day. This nonspecific, general endurance is somewhat a function of aerobic capacity ($\text{VO}_2 \text{ max}$), but it's largely influenced by available fuel supply (resident glycogen stores and carbohydrate consumption during exercise) and mental toughness (the will to endure pain and reach your goal).

Muscular endurance is the capability to sustain high-intensity anaerobic activity for a minute or more, or to maintain a high power output through several bouts of intense activity with only brief rests breaks in between. Failure on a near-limit rock climb often seems to come down to a lack of local forearm muscle endurance, as the hallmark muscle pump and burn seems to mandate an end to your effort. Local muscular endurance is a function of your level of maximum strength, your muscles' ability to tolerate acidosis, energy substrate availability, and your will to sustain strenuous activity despite painful stress signals from the brain.

Per the Central Governor Model described above, the brain limits power output and creates pain sensations and stopping thoughts so that you will terminate activity long before you risk catastrophic injury or death. Effective training, then, must not only address specific physiological constraints (aerobic and anaerobic capacity), but also train the brain to better interpret afferent feedback from the blood, muscles, and other organs and thus increase the brain's threshold for shutting down muscular activation during strenuous exercise. The key is to train in ways that closely simulate the demands of your goal climbs, and in doing so to stretch your limits in anaerobic and aerobic endurance to establish new central governor experience and fatigue set points.

In closing, top climbers—like Adam Ondra, Tommy Caldwell, Chris Sharma, Daniel Woods, and Alex Megos—have spent years training and climbing at progressively higher intensities and for longer durations. Through a combination of deep commitment to training and exercising incredible willpower, they have stretched their brains and bodies to new levels. While you and I may not possess quite as good genetics (or the time and resources) of these pros, we undoubtedly possess room to improve, both physiologically and in terms of recalibrating our central governor.

Research on Rock Climbing—The First 2,000 Years

The earliest sports science research on rock climbing primarily focused on specific sites and mechanisms of injury. The development of organized difficulty-format competitions at the World Cup level in the 1980s encouraged exploration of the climber as an athlete. Initially, scientific study focused on defining the performance model for difficult climbing—the identification of measurable factors that seemed to be common in elite climbers. Following pathways of discovery that were typical in other sports, these initial studies focused on anthropometry, or stature and body composition; muscular strength; and energy expenditure.

The first large-scale anthropometry study of elite competition climbers was published by Watts, Martin, and Durtschi (USA) in 1993. In a sample of thirty-nine international competitors, the athletes were found to be small in stature and low in body weight with very low body fat levels. It is possible this early observation of very light and extremely lean climbers at the elite level encouraged aspiring competitors to reduce body mass through special diets and borderline disordered eating. Today we know severe reductions in body weight, and body fat in particular, can negatively impact maintenance of muscular strength and health in general. With this knowledge, the dominant anthropometry of elite climbers has likely changed a bit since the 1980s. Climbing steeper, more physically demanding terrain may also be a factor in some climber anthropometry characteristics today.

Many early studies focused on strength and found elite climbers to have relatively average handgrip strength when compared with gender- and age-matched population norms. However, when grip-strength values were expressed relative to body weight, elite climbers scored in the top 10 to 20 percent. Subsequent study and observation of the hands during climbing revealed minimal use of the standard handgrip testing position where the thumb applies

force in opposition to the fingers. Researchers such as Grant (Scotland), Watts (USA), Schweizer (Switzerland), Vigoroux (France), and others developed new methods to test finger flexion force through force transducers mounted in adjustable frames. Such research found climbers to be superior in sustaining repetitive contractions with the muscles that control finger force and specific hand configurations.

A developing area of research is the rate of force development (RFD) in the finger flexor muscles. Fanchini (Italy), along with coworkers in Switzerland and France, found RFD to be high in boulderers and lead climbers relative to non-climbers. Interestingly, the boulderers had significantly higher RFD than the lead climbers. There is room for more research in this area.

Muscle endurance has also been of interest. New studies by Fryer and coworkers (UK and Europe) and others have looked at factors that affect muscle contraction endurance in the hands and recovery between repeated contractions. Higher forearm blood flow and better oxygenation of forearm muscles have been found in elite climbers. These factors likely improve the climber's resistance to fatigue throughout the ascent of a route.

Bioenergetics—energy expenditure and the support systems for energy expenditure—has also been of interest in early research and continues to be explored. Billat (France) and coworkers were the first to report oxygen uptake (VO_2) values for expert climbers during climbing. Watts and Drobish used an early nonmotorized climbing treadmill (Brewer's Ledge Treadwall) to study VO_2 and energy expenditure during continuous climbing at different angles. Development of battery-powered portable metabolic analysis systems in the late 1990s and 2000s opened up the study of bioenergetics during actual climbing on indoor and outdoor terrain. Although results were specific to test protocol, climber ability level, and

speed of climbing, the oxygen uptake rate during sustained climbing seemed to range between twenty-five to thirty-five milliliters of oxygen per kilogram of body weight per minute with energy expenditures of around eleven kilocalories per minute. While these studies illustrate the aerobic nature of climbing, other studies have found elevations of lactate in the blood, which can be a marker for anaerobically derived energy. Bertuzzi and coworkers (Brazil) have estimated energy to be derived 42 percent through aerobic metabolism, 22 percent from an anaerobic lactate-producing process, and 36 percent from breakdown of high-energy phosphates without lactate production during difficult climbing. Thus, climbing involves a broad range of bioenergetics.

Taken as a whole, the historical and more contemporary physiological research on difficult rock climbing yields a complex portrait—strength with endurance, peak muscular effort and rate of force development, aerobic energy expenditure with high-intensity power, explosiveness with subtle control. Researchers such as Jiří Baláš (Czech Republic) and Vanesa España-Romero (Spain), along with their coworkers, continue to pursue factors in the physiological performance model

for climbing. Peter Wolf and coworkers in Switzerland have begun research with very sophisticated instrumented modular holds that should yield new information regarding the forces involved in different climbing moves and techniques.

The era of research for rock climbing is relatively young. Although the background physiological model has been refined, a lack of controlled training research requires careful application of known principles and established training theory based upon the model. Fortunately, *Training for Climbing* does this. New investigators, such as López-Rivera and González-Badillo (Spain), are starting to study climbing-specific training protocols. It is an exciting time for the science of rock climbing.

This brief overview of research does not even begin to explore the associated areas of biomechanics, nutrition, and psychological factors! Thankfully, Eric Hörst follows the contemporary research intensely and incorporates the best of our current knowledge into his website and books. Another valuable resource for tracking research is the International Rock Climbing Research Association (www.ircra.rocks).

—Dr. Phil Watts

Training Principles

Backed by the most rigorous exercise science, we can now embark on a quick study of nine training principles. Some of the nine principles presented below will sound familiar to you, while others may seem novel. Trust me, however, that they are all powerful! You might consider these as simply guideposts to effective training and exercise programming. The most pragmatic climbers—like this author!—will fastidiously abide by the principles with the goal of squeezing the maximum output in performance gains for the time-limited training input. So let's get started with the principle of specificity, the paramount of all training principles.

Specificity

The principle of specificity of training simply states that the more specific a training activity is to a given sport—in direction of force application, velocity of movement, pattern of movement, body posture, range of motion, type of contraction, and energy systems used—the more it will contribute to increasing performance in that sport. Therefore, for an exercise to be effective at producing usable strength gains for climbing (such as grip strength, lock-off strength, or lunging power), it must impart a stress on the musculature in a very climbing-specific way. The more specific the training activity or exercise, the greater the benefit on your climbing

performance. Let's look at a few examples of how this rule applies to training for climbing.

Circuit training or pumping iron does not train the muscles in the slightest way similar to their use in rock climbing. Consequently, doing a health club-style workout (using machines and free weights exclusively) is largely a waste of energy in terms of improving climbing performance (the exception being individuals possessing unusually poor levels of general fitness). A better approach, according to the principle of specificity, would be to climb three days per week and design any supplemental weight training to target the important pulling and antagonist muscles, as well as the core and extension muscles of the back and legs.

Squeezing a rubber hand doughnut (or other similar spring-loaded device) is likewise unproductive for improving your finger strength for climbing. Grip strength shows a remarkable amount of specificity depending on the grip position (crimp, open hand, pinch), the positions of the wrist and elbow, the intensity of the contraction, and even the type of contraction (isometric or concentric). Furthermore, since your grip tends to fail while you are pulling down on it with near-maximum load, it must be trained in much the same way. Consequently, squeezing a rubber doughnut is basically useless as climbing training, though it does have some value as a warm-up exercise and in injury rehabilitation.

What about the basic pull-up (palms away), a most popular exercise among climbers? Obviously the motion is similar to climbing, but your posture, your degree of body tension, and the exact positions of your hands and arms do not vary randomly as they do on rock. What's more, the ability to stop or lock off your arm in some novel position is often more vital in climbing than is the simple act of pulling. Therefore, to produce the most transfer of your pull-up training to the rocks (and reduce your risk of an elbow overuse injury), you want to alter the pull-up in a variety of ways with every set. For

example, you might change the distance between your hands, stagger one hand lower than the other (use a webbing loop), and include some lock-offs or stops in the motion at a variety of arm angles. This approach would be much more advantageous than just doing pull-ups in the same fixed position.

Finally, let's consider the concept of cross-training as some individuals try to apply it to climbing. Clearly, the idea that performing any other sports activity might improve climbing performance is in blatant conflict with the principle of specificity. In fact, the only sports in which cross-training seems to be practical are the aerobic endurance sports, as popularized by the triathlon phenomenon.

Individualization

This could also be called the snowflake principle, since it highlights that no two climbers—or their optimal conditioning programs—are the same. The best training program for you will target your specific weaknesses, address past or present injuries, provide sufficient time for recovery, and be structured to provide the greatest output for the available training input. Since there is no other climber quite like you, there is no other climber's training program that you'd want to copy—doing so will provide less-than-optimal results and might even get you injured. Consequently, it would be wise to develop and execute what seems to be the best program for you and ignore how others train. As a guiding principle remember that “no one is you, and that's your power!”

Progressive Overload

This granddaddy of training principles states that in order to increase physical capability, it is necessary to expose your body to a level of stress beyond that to which it is accustomed. You can achieve this overload by increasing the intensity (greater resistance), volume, or speed of training, or by decreasing the rest interval between successive sets. Depending on the exercise and which of these

exercise parameters you choose to vary, the overload will effect adaptations resulting in more strength, power, anaerobic endurance, or stamina. For example, increasing exercise resistance and speed will produce gains in maximum strength and power, whereas decreasing rest intervals and increasing volume will improve muscular endurance.

While it's probably a good idea to vary the method of overload from time to time, the best method of overload for you depends on your climbing preference. If bouldering is your favorite type of climbing, then you'd want to favor training that builds strength and power (and create overload by increasing resistance and recruitment). In training for roped climbing, however, it would be best to increase volume and decrease rest intervals to improve muscular endurance. Finally, big-wall and alpine climbers looking for greater stamina should create overload by increasing total volume of moderate climbing and generalized aerobic activity.

Variation

One of the most common training errors among all athletes is the failure to regularly change their training program. This principle states that the body becomes accustomed to training stimuli that are repeatedly applied in the same way. Therefore, if you go to the climbing gym and engage in the same basic routine every time, your strength and climbing gains will eventually plateau despite what feel like good workouts. Strive to vary your training by manipulating the type of overload (per above) as well as mixing up the type and order of climbs and exercises performed.

Periodization, another form of variation, involves alternating the overall workout intensity and volume over the course of days or weeks. For example, with indoor training you might alternate workouts among “high volume” (doing many moderate routes), “high intensity” (hard, powerful bouldering), and “high, high” (climbing several near-limit routes). You could also vary your

workouts every few weeks as in the 3-2-1 training cycle described in chapter 10. Bottom line: Make the Principle of Variation a cornerstone of your training-for-climbing program and you will get uncommonly good results!

Isolation

While beginning climbers typically experience steady gains in climbing ability and climbing-specific strength for up to two or three years, reaching a performance plateau is exceedingly common (often around the 5.11 or 5.12 grades). Breaking through this frustrating ceiling often requires embarking on a new training program that incorporates highly specific exercises that target the limiting constraints—most commonly, finger flexor strength and endurance, limit strength and power in the pulling muscles, and the ability to create core stiffness.

Applied literally, the principle of isolation requires targeting a single muscle group while activity in all nearby or related muscles is minimized. In reality this is very difficult to do, especially when you want to obey the principle of specificity by isolating a muscle in a climbing-specific way. Still, climbers can exercise in the spirit of this principle—and get excellent results—with quasi-isolation exercises such as fingerboard hangs, certain forms of campus training, System Wall “isolations,” and various core-muscle exercises. Such isolation training provides tremendous stimulus for neural and morphological adaptations, by maximizing fiber recruitment and energy system efficiency.

Optimal Sets, Reps, and Rest Intervals Between Sets

The optimal training load (resistance), the number of reps and sets, and the rest interval between sets depend on the desired training outcome. Warm-up sets and stabilizer-muscle training (e.g., rotator cuff, wrist extensions, and scapular stabilizers) are best performed with a relatively modest resistance that allows fifteen to thirty repetitions. Similarly,

antagonist-muscle exercises need only be performed at a moderately-high resistance with ten to twenty reps per set. For the purposes of developing climbing-specific strength and local endurance, however, a much higher workload (in some cases five to ten sets) is needed to produce beneficial results on the rock.

The length of rest between sets can similarly be managed to produce optimal training adaptations. A three- to five-minute recovery period is best when training for strength and power. A much shorter rest period—thirty seconds to two minutes—is optimal when training for improved local endurance. No need to be long-winded here, since chapters 8 and 9 will provide dozens of exercises and outline the best rep and set scheme and recovery interval for each.

Exercise Order

Every workout should, of course, begin with warm-up period that is gradually progressive. Once all the muscles have been warmed and “turned on,” the next segment of your workout should involve actual climbing—practicing new moves, working hard boulders, rope climbing, and such—since being mentally and physically fresh is important for developing and reinforcing efficient motor and movement skills. Only upon completion of your climbing should you move on to targeted supplemental exercises such as weighted pull-ups, fingerboard hangs, campus training, and the like. Follow these activities with stabilizer and core training, and conclude with some foam rolling and stretching to jump-start the recovery process.

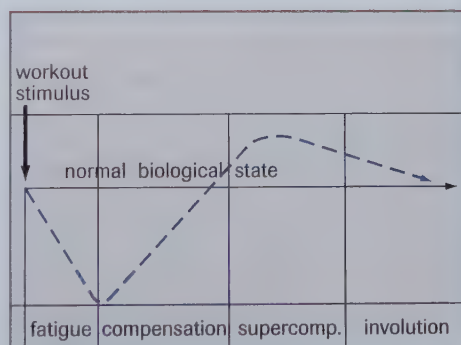
Any extensive antagonist-muscle training, supplemental free weight work, or aerobic exercise is best performed on a rest day from climbing. As you’ll learn in chapter 6, there’s no need to train the antagonists more than twice per week. Some climbers, however, may wish to do aerobic training two to four days per week, while also climbing three or four days per week and taking one full day of rest per week. The only way to schedule all this (and

get one day of complete rest per week) is to do a couple of split workout days—that is, days in which you do a morning and late-day workout. The most effective way to split workouts is to do an aerobic session as early in the morning as possible, and then the climbing workout in the late afternoon or evening. Scheduling six to eight hours rest between the split sessions is essential, and you should always train general endurance early and strength/power endurance late in the day—you’ll get suboptimal results the other way around.

Rest

The various muscular adaptations described earlier occur between, not during, workouts. Sufficient rest and healthy lifestyle habits (including proper nutrition and adequate sleep) are fundamental to maximizing the adaptations from your training investments. As a rough guideline, complete recovery (supercompensation) takes anywhere from twenty-four to seventy-two hours, depending on the intensity and volume of the stimulus (see figure 5.5). For example, it might only take one day to recover from a high volume of low-intensity activity

Figure 5.5 Supercompensation Cycle



Complete recovery (supercompensation) takes anywhere from twenty-four to seventy-two hours, depending on the intensity and volume of the training stimulus.

like climbing a bunch of relatively easy routes or just hiking, whereas it will likely take two or three days to recover completely from a high volume of high-intensity exercise, such as climbing many routes near your limit, or performing a rigorous strength-training session with hypergravity pull-ups, weighted fingerboard hangs, campus training, and such.

The importance of this principle cannot be overstated, since training too often (under-resting) will eventually lead to a decline in performance and/or injury (see figure 5.6). This is known as the overtraining syndrome, and it's surprisingly common among passionate climbers. Observe how many climbers out there are whining about their nagging injuries or how they are “not getting stronger” despite their dedication to rigorous training. That's why it's called overtraining!

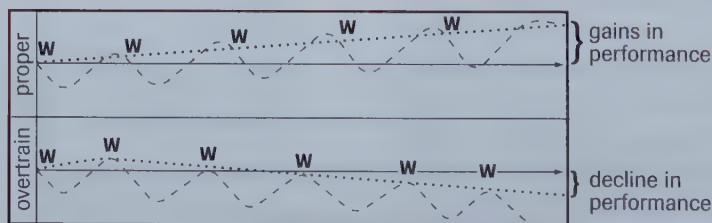
Another factor leading to overtraining or unusually long workout recovery is the frequent tendency of doing too high a volume of high-intensity exercise in a single session. As shown in the Supercompensation Cycle (figure 5.5), the workout stimulus results in neuromuscular fatigue and a temporary degradation in functional ability. With adequate rest the system regenerates to a level higher than before the workout. Too hard of a workout, however, creates so much neural fatigue and cellular damage that it takes much longer to recover and realize a state of supercompensation. This is an important concept to keep in mind when

performing high-intensity training. Doing twelve sets of campus training probably provides no more beneficial stimulus than six or eight sets, but by doing twelve sets you dig a deeper hole from which it will take longer to recover. The same argument could be made against doing thirty sets of pull-ups or spending sixty minutes training on a fingerboard. Summing up: When it comes to high-intensity training, less is often more.

Detraining

Upon cessation of strength or endurance training (or frequent climbing), recent functional gains begin to erode in as little as seven days of rest. Local endurance diminishes more quickly than strength, and noticeable loss of endurance is likely after just seven to ten days away from climbing. Research has shown that half the mitochondria (and aerobic enzyme) gains from four weeks of training can be lost in as little as a week, and all is lost in five weeks of inactivity. Regaining the one-week loss of mitochondria content will take up to four weeks of endurance training. Realizing the remarkable plasticity of mitochondria—and how quick you'll lose your endurance—should be enough to compel you to go climbing once or twice per week! Fortunately, strength gains don't diminish as quickly—two weeks away from climbing or the gym may yield no remarkable loss in strength, although slacking much longer than this surely will.

Figure 5.6 Long-Term Training Response



Adequate rest between workouts allows full supercompensation and long-term gains in performance. Conversely, executing workouts too close together results in a long-term decline in performance.

Tips for Effective Physical Training

1. Train specifically for climbing! While supplemental training of the antagonist and stabilizer muscles is essential, climbing-specific training holds the greatest potential to improve your climbing performance.
2. Target your limiting constraints. There are no other climbers quite like you, so performing some other climber's training program will be less than optimal. Identify what's holding you back (technically, physically, and mentally) and get busy training to improve in these areas.
3. Employ progressive overload into your weekly training. While not every workout should be the "hardest ever," it is important to occasionally expose your body to a level of training stress. This can be achieved by increasing training intensity, speed, and volume, or by decreasing the rest interval between sets or climbs.
4. Vary your workouts. Since the body adapts to training stimuli, it's essential to regularly vary training activities and workouts every few days or weeks, using either autoregulation or periodization.
5. Use muscle isolation training to break through plateaus. High-level climbers must employ isolation training to eke out small gains in strength, endurance, and power.
6. Use the optimal set, rep, and rest scheme to produce the desired gains. Write down your workouts in order to keep track of—and properly adjust—your weight, rep, and rest scheme to obtain optimum results.
7. Arrange your workouts from specific to general. Always train climbing movements (actual climbing) first, followed by climbing-specific exercises and then general strength training. Save extensive aerobic activity for a non-climbing day, or perform such training early in the day (and climb late).
8. Plan your rest days just as you plan training days. Neuromuscular adaptations occur during periods of rest and sleep, not during workouts. Therefore, sufficient rest and a healthy lifestyle are essential for making the most of your training investment.
9. Avoid long breaks from training or climbing. Taking more than a week or two off (from training or climbing) will result in a significant loss of climbing-specific endurance and strength. In other words, if you don't use it you will lose it!

Training Methodology

I'll close out this chapter with a brief, targeted look at the methodology of training strength, power, local endurance, and stamina. Founded on the exercise physiology and principles outlined throughout this chapter, the methods will then come to life in the exercises and training protocols detailed in chapters 6 through 9.

Maximum-Strength Training

In climbing, your level of maximum strength in the forearm flexor and pulling muscles (of the arms and torso) is a common physical limitation. Gripping small holds, making a powerful arm pull,

and locking off with one arm all command your muscles to contract briefly with near-maximal force. Interestingly, your ability to rest effectively on a barely adequate, medium-size hold and express strength-endurance on a pump sequence is also a function of your maximum strength. Therefore, training for greater absolute grip and pulling strength is the single most important aspect of an effective training-for-climbing program.

So what is the best way to train this all-important attribute? For beginning climbers, simply climbing three days per week will yield some gains in climbing-specific strength. Therefore, no highly targeted training is necessary, or appropriate—it could very well lead to injury. Of course, training

technique and movement skills are paramount at this stage, and any time spent strength training should be focused on the antagonist, stabilizer, and larger pulling muscles (not the fingers).

Intermediate climbers, with at least a couple of years of climbing experience, will benefit greatly from targeted strength training—and having presumably developed enough tendon strength, these mid-level climbers can begin a progressive twice-per-week strength-training program. Elite climbers are in a class of their own: With years of climbing experience and (apparently healthy) Kevlar-like tendons, these elites can embark on an ultra-intense, two- or three-day-per-week strength- and power-training regimen. Breaking through the next performance plateau depends on it.

As you learned earlier in this chapter, attaining a higher level of maximum strength is a matter of increasing neural recruitment, muscle hypertrophy, and building greater cellular stores of ATP-CP. The training goal is to maximize fiber recruitment and liberate ATP-CP at the highest possible rate for five to ten seconds. Consequently, a properly executed maximum-strength exercise will utilize a sufficiently high resistance to produce near-failure in around ten seconds. It's important to recognize that any strenuous exercise performed for more than about twelve seconds will train local endurance more than

strength, as these longer efforts are fueled more by the anaerobic lactic system. Table 5.4 shows the ideal exercise duration, intensity, number of sets, rest interval between sets, and frequency of use. Chapters 8 and 9 will provide details on many strength-training exercises, but here are a few examples:

- Weighted pull-ups with enough added weight to make five repetitions difficult.
- Body weight fingerboard hangs on holds small enough to be difficult if held for ten seconds.
- Weighted fingerboard hangs on medium-size holds with enough weight to make a ten-second hang very difficult.
- One-arm lock-offs held for five seconds.
- One-arm (or one-arm-assisted) pull-ups for one to five repetitions.

A final note: Near-limit bouldering can aid in the development of maximum strength, especially in sub-elite climbers; however, more highly targeted supplemental exercises are essential to provide optimal stimuli for maximum-strength gains. Consider that failure on boulders often occurs because of movement flaws, inadequate flexibility, or lack of power, and therefore bouldering does not necessarily elicit grip or pull-muscle failure in less than twelve seconds. Shrewd, precise training—in

Table 5.4 Weight, Rep, and Rest Interval Schemes

	Intensity (% of 1RM)	Reps	Duration	Sets	Rest Interval	Frequency
Maximum Strength	90–100%	3–5	<12 secs.	3–5	3–5 mins.	2x per week
Strength-Endurance	80–90%	5–8	12–30 secs.	5–10	1–3 mins.	2x per week
Hypertrophy	70–80%	8–15	20–40 secs.	3–5	2–3 mins.	3x per week
"Local" Aerobic Endurance	50–70%	>20	>60 secs.	1–5	1–3 mins.	2–3x per week

Use these guidelines to get the best response from your strength and endurance training. Note that you may be able to avoid unwanted muscle mass gains by eschewing hypertrophy training (highlighted).

accordance with the exercise science and principles detailed in this chapter—is paramount for eking out additional strength gains over the long term. Train smarter to climb harder!

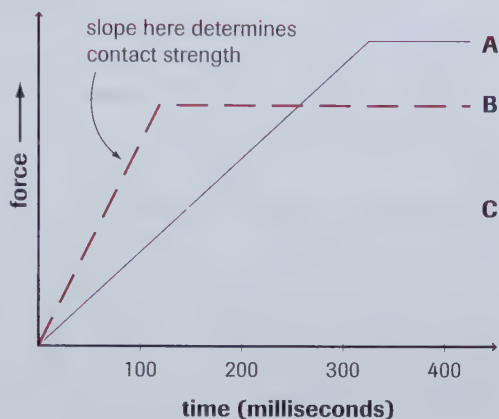
Contact Strength and Power Training

When climbers talk about “power,” they typically are referring to the ability to make a swift strenuous reach, campus move, or quick latch on to a small handhold at the end of a deadpoint or lunge move. Physiologically, power is strength expressed quickly (power = strength x speed), so your level of power is closely related to your maximum strength. Both power and maximum strength involve forceful fast-twitch fiber recruitment, and both are fueled by cellular ATP-CP, which can only sustain near-maximal power (or strength) for a few fleeting seconds. Where power and strength significantly differ is in the rate at which a force is applied—sports scientists call this rate of force development (RFD).

To understand RFD, consider this on-the-rock example of how power and strength are expressed differently in grabbing on to a difficult handhold. Figure 5.7 shows hypothetical force–time curves for three climbers. Climber A possesses the greatest grip strength and can hang on very small holds, given a second or so to summon his peak strength. Climber B has less absolute strength than Climber A, but she is more powerful, because she can summon her strength more quickly (greater contact strength)—ideal for catching dynos and latching small holds when deadpointing. Climber C is neither strong nor powerful—he needs to begin a strength-training program; otherwise he’d better stick to climbing slabs!

Obviously the goal should be to maximize your strength and power, much like Climber B. Possessing a high absolute strength and the ability to recruit this strength quickly is essential for climbing hard boulders and powering through short, burly cruxes on roped climbs. While the best exercises and sample training programs will

Figure 5.7 Rate of Force Production



Hypothetical grip strength of three climbers. Climber A has the greatest absolute grip strength, while Climber B has the best contact strength.

be detailed in later chapters, one fundamental rule is to “develop strength before power.” Therefore, no matter your current level of pulling power or contact strength, beginning a new training cycle should first focus on maximum-strength training (for a few weeks) before progressing to a mix of strength and power training.

Effective contact strength and power-training exercises must elicit extremely rapid recruitment (but always in a semi-controlled way) and focus on maximal (anaerobic) power output for one to ten seconds—continuing any longer than about twelve seconds shifts the focus to anaerobic lactic capacity, which is something completely different (see sidebar on “Energy Systems Training”). Here are a few of the many power-training exercises you’ll learn in chapters 8 and 9:

- Clap pull-ups
- No-feet gym rope climbing
- Feet-off campus laddering
- Campus Board “Switch Hands”
- Double dyno campus training

Energy Systems Training

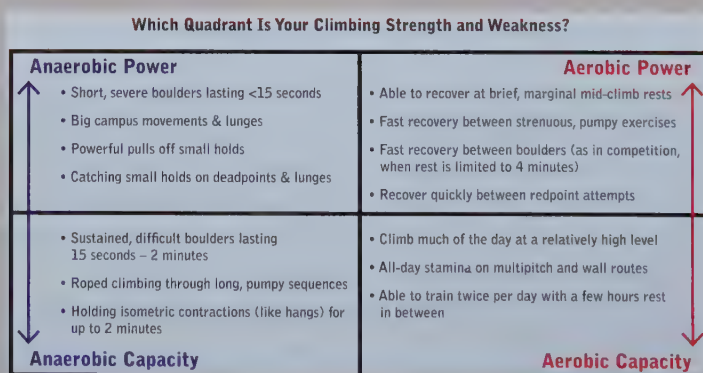
While all climbs—short or long and slab or steep—use all three energy systems simultaneously, each energy system has specific situations in which it dominates (as described on page 94). Therefore, the relative strength of your energy systems will explain some of what you experience as your physical strengths and weaknesses on the rock. While a good coach can accurately assess the state of your three energy systems, you can glean a subjective sense of where you stand with each based on the energy system characteristics shown in Figure 5.8.

A petrol-fueled automobile provides a good analogy to better understand how the energy systems apply to climbing. Think of “power” as the size of the engine—the anaerobic power system is like an eight-cylinder engine, whereas the aerobic power system is like a small four-cylinder engine. If you are climbing an extremely strenuous sequence, you need V8-like power output that only the anaerobic alactic system can provide. Much less intense climbing, say exerting only 40 percent of peak power on a climb a few grades below

your limit, can readily be fueled by the less-powerful aerobic energy system.

Staying with the car analogy, the “capacity” component is like the size of the gas tank; through proper training you can increase both anaerobic and aerobic capacity (the size of each tank). It’s important to understand, however, that the anaerobic “tank” will always be but a small percentage of the size of the aerobic tank; furthermore, aerobic capacity is far more trainable than anaerobic capacity. Therefore, if you want to climb a lot in a given session or day—many boulders, sport routes, or all-day gear climbing—you need a well-developed aerobic system. For boulderers and sport climbers, training aerobic power is extremely important and often overlooked. As explained earlier, it’s the aerobic system that resynthesizes CP (for quick recovery of anaerobic power) and helps clean up the metabolic by-products of anaerobic glycolysis (thus refueling your anaerobic “tank” during periods of rest). This is a critical distinction that all serious climbers should recognize and apply to their training.

Figure 5.8 Energy System Characteristics—What’s in Your Wheelhouse?



Considering these energy characteristics, which quadrant seems to be the locus of your physical strength (and weakness) on the rock?

Table 5.5 Summary of Energy Systems and How They Power Your Climbing

Anaerobic Power	Aerobic Power
Role: Very high power output for less than 12 seconds. No muscle pump.	Role: Power low to moderate intensity exercise. Facilitates rapid recovery between bouts of anaerobic exercise.
Energy System: Anaerobic alactic	Energy System: Aerobic
Fuel: Cellular ATP-CP	Fuel: Glycogen, lactate, fatty acids
Most Important for: Bouldering, power moves, maximum strength, physical crux moves	Most Important for: Regenerating ATP-CP for the anaerobic alactic system, clearing excessive lactate accumulation due to anaerobic glycolysis.
How to Train: Maximum-strength exercises, brief maximal-power exercises, alactic intervals of less than 15 seconds (no pump).	How to Train: Alactic intervals of less than 15 seconds, Strength/power-endurance intervals of 15 to 90 seconds.
Adaptations: Greater recruitment and increased cellular ATP-CP storage.	Adaptations: Increased mitochondria efficiency and capillarity, increased aerobic enzymes, increased cardiac output and respiration.
Anaerobic Capacity	Aerobic Capacity
Role: High power output for 12 seconds to 2 minutes. Results in pumped, burning muscles; shortness of breath.	Role: Provides long-duration, low to moderate power output lasting many minutes to hours.
Energy System: Anaerobic lactic (glycolysis)	Energy System: Aerobic
Fuel: Muscle glycogen	Fuel: Glycogen, lactate, fatty acids
Most Important for: Long boulders and cruxes, hard sustained sequences lasting up to 2 minutes without rest.	Most Important for: Local endurance of more than 2 minutes, long climbs, and all-day stamina.
How to Train: Exercises and climbing intervals lasting 30 to 90 seconds with 1:4 to 1:6 work-rest ratio.	How to Train: Threshold intervals of 2 to 5 minutes with short rests (1:1 or 1:2 work-rest ratio), sustained easy to moderate climbing of 5 to 20 minutes, unspecific aerobic training (running, etc.)
Adaptations: Improved buffering and tolerance to acidosis, increased cellular glycogen storage, increased anaerobic enzymes, increased mitochondrial and capillary density.	Adaptations: Increased VO_2 max, increased cardiac output and respiration, increased liver glycogen storage, increased aerobic enzymes, increased mitochondrial and capillary density.

Strength-Endurance and Power-Endurance

Your strength-endurance and power-endurance are a direct function of your anaerobic capacity (see the “Energy Systems Training” sidebar). Training the anaerobic lactic energy system is painful and grueling, but doing the right amount of this type of training is extremely important if your climbing goals include sending difficult rope-length routes or long, sustained boulder problems.

As shown in Table 5.3 and Figure 5.3, true maximum strength and power output can only be sustained for a few fleeting moments. Strength- and power-endurance, then, relate to how long the muscles can produce high, but submaximal, power output despite frequently occluded or partially occluded local blood flow and the resultant increase in hypoxia and metabolic acidosis. A rapid decrease in anaerobic energy production will take place between thirty seconds and two minutes, depending on power output levels and a person’s anaerobic capacity. Individuals with high anaerobic capacity will excel at climbing long, powerful sequences—despite a growing pump—lasting upward of two minutes without a rest, whereas a climber of similar technical ability but lesser anaerobic capacity will fail more quickly, perhaps in less than one minute.

Another important distinction relates to how long a climber can persist on a hard, sustained route with frequent brief, marginal rests. Possessing a high aerobic power will facilitate meaningful recovery in as little as five to ten seconds of shaking out at a marginal rest position. Thus, in watching a climber hang on through a long, powerful route, it’s often hard to tell if they are prevailing due to a high anaerobic capacity or because they are able to benefit from very brief, marginal rests thanks to a high aerobic power contribution. Ultimately, you want to train both systems, since having high aerobic power speeds regeneration of ATP-CP and helps “clean up” the metabolic by-products of the anaerobic system.

Your anaerobic capacity is expressed on the rock in both your endurance of forearm strength (how

long you can grip hard) and your power-endurance in pulling hard, deadpointing, and lunging. Therefore, you need to train anaerobic endurance using both static exercises (isometrics) and powerful movements. Furthermore, you must train using both longer-duration exercises (that significantly occlude blood flow) and short bursts of high-intensity interval training that allows blood to flow briefly between exercise bouts. Here’s a sampling of the strength-endurance and power-endurance exercises you’ll learn about in chapters 8 and 9:

- 20-20 fingerboard repeaters
- Pull-up Interval (10 to 20 sets of 5 to 10 reps)
- 1-minute climbing intervals (10 to 20 sets with a 1-minute rest between)
- Bouldering 4x4s
- Campus Board “Ladder Laps”
- HIT System training (strength-endurance protocol)
- Route intervals (2 minutes sustained climbing with 6 to 10 minutes of rest)

Proper scheduling of anaerobic lactic energy system training is just as important as the exercise choice. While anaerobic endurance training will yield noticeable gains in as little as two weeks, the anaerobic lactic energy system is arguably the least trainable of the three energy systems. Therefore, training gains usually cease in about four weeks, at which time the training focus should be temporarily shifted to another energy system. Another critical scheduling point—unknown by many coaches and most climbers—is that doing too much anaerobic endurance training (more than two or three sessions per week) will eventually lead to decreased endurance due to the damage chronic acidosis does to anaerobic enzymes and mitochondrial function. If you frequently train to the point of a deep, burning pump yet feel you’re not gaining any endurance on the rock, you now know why.

What's More Important, Strength Training or Endurance Training?

Maximum strength, power, and contact strength (grip) are undoubtedly essential for bouldering at a high level. A recent study confirms this notion, showing that boulderers' crimp and open-hand strength was about 12 percent stronger than roped climbers. Even more remarkable, the boulderers' rate of force development in the finger flexors (contact strength) averaged 37 percent faster than roped climbers (Maurizio 2013). Interestingly, a study comparing advanced and elite rope climbers found that climbing time to exhaustion was the factor that most determined performance and on-sight level (España-Romero 2009).

These studies confirm the intuitive conclusion that boulderers should favor strength and power training, while roped climbers must focus on improving both anaerobic and aerobic endurance. But the science of climbing isn't so straightforward, and you won't get optimal long-term results by training in just one way. Earlier in this chapter I mentioned a study showing how

endurance-trained climbers recovered more quickly between climbs. Another climbing study (Grant 2003) found that finger endurance (time to failure) is essentially fixed when holding an isometric grip at 40 percent of maximum voluntary contraction (MVC). Thus, increasing absolute finger strength (MVC) will yield meaningful endurance gains when carrying submaximal amounts of body weight. Another way to interpret this subtle but powerful study is that a stronger climber can rest (successfully) on smaller holds—important for pushing the limits on roped climbs!

The bottom line: While boulderers must concentrate on strength and power training, it would be beneficial to perform one or two endurance-oriented sessions per week. Similarly, a roped climber can benefit greatly from frequent strength and power training in addition to their steady commitment to training local aerobic and anaerobic capacity. See chapter 10 for specific training schedules for both boulderers and roped climbers.

Climbing Stamina and Nonspecific Aerobic Training

Stamina training and strength training are at opposite ends of the training spectrum. Fast, brief, anaerobic-powered movements are fueled by cellular ATP-CP, while the slow, steady, high-capacity aerobic system is fueled by fatty acids, lactate, and glycogen. Furthermore, powerful anaerobic movements call a large portion of FT fibers into action, whereas ST fibers dominate during lower-intensity, sustained activities. As you might expect, then, the methodology of effective stamina training is vastly different from that of training for absolute strength and power.

Effective stamina training necessarily involves a much higher volume of climbing (and aerobic

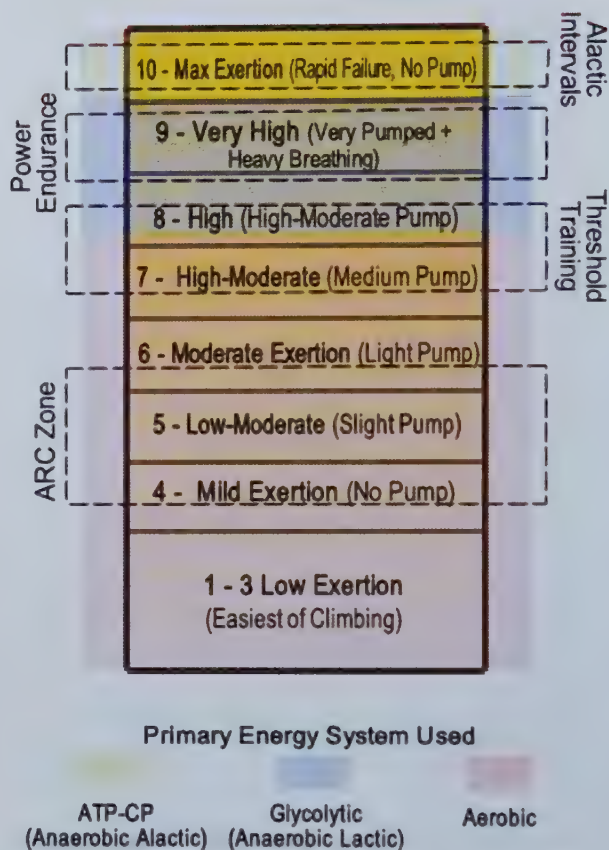
exercise) than when engaging in strength or power training. In order to deeply train the aerobic energy system, it's essential to spend more time actually exercising rather than resting (the opposite is the case when strength training). Naturally, the exercise intensity of these stamina sessions is much lower and primarily targets the ST and FTa oxidative muscle fibers. No matter your choice of climbing-specific or nonspecific exercises, the ideal stamina training zone is 50 to 80 percent of maximum intensity—you can best hit this target zone by aiming for a subjective effort level (intensity) of between 5 and 8 on a scale of 1 to 10. Let's take a closer look at climbing-specific and nonspecific aerobic energy system training.

CLIMBING-SPECIFIC AEROBIC TRAINING

Effective training of the aerobic energy pathway via climbing is only possible if you set aside the desire to climb for performance (near your limit) and instead embrace the potentially boring process of climbing high volumes of moderate rock (or plastic). No matter if the climbing is done indoors or outside, this approach will naturally involve lots of submaximal climbing at varying intensities. The critical guideline to obey is that the climbing intensity never ascends deeply into the anaerobic zone, the hallmark of which is a deep muscle pump and shortness of breath. Ultimately, you want to find the margin of the anaerobic lactic zone and strive to climb mostly just below it (in the aerobic zone) while only occasionally crossing into the low end of the anaerobic zone. A light to moderate forearm pump is fine and desirable; however, a flaming pump, heavy breathing, and a growing sense of losing control is a clear sign you've climbed too deep into the anaerobic zone.

Interestingly, I've observed that many climbers who set out to train climbing stamina in this way ultimately end up crossing the line and getting massively pumped (thus training the anaerobic lactic system rather than the aerobic system as intended). It seems that wanting to get pumped while training is in the DNA of most climbers, yet this "pump lust" must be denied when training the aerobic energy pathway. Self-control and a strong intention to climb mainly in the aerobic training zone are essential to properly training this energy pathway. By keeping your perceived exertion and exercise intensity between 5 and 8 (see figure 5.9), you can rest assured that you're training the aerobic energy system. Threshold intervals (as detailed in chapter 8) requires a steady climbing exertion of 7 to 8.5 out of 10, whereas recovery "ARC training" demands you limit exertion to just a 4 to 6 out of 10.

Figure 5.9 Training Zones for Climbers



Energy system training zones for rock climbers based on subjective feeling of forearm muscle pump and perceived exertion. Effective climbing-specific aerobic training must be performed between 5 and 8.5 on the rating of perceived exertion (RPE).

Here's a preview of a few climbing-specific exercise strategies—all should be performed in the 5 to 8 training zone.

- Moderate bouldering traverses (indoors or outside) with twenty to sixty minutes of total climbing time.
- “Threshold” intervals on a Treadwall or bouldering wall, three to six minutes with near equal-length rests for a total of twenty to thirty minutes of climbing time. Strive to climb at an intensity of around 8 out of 10 on the RPE scale.
- Climb many moderate roped routes (toprope or sport lead), being sure to never get more than a moderate pump. Err on the side of selecting routes that are too easy rather than too hard.
- Long days of outdoor climbing—the goal is to get fatigued but not pumped. Doing eight to fifteen pitches that are two to four number grades below your limit is a good goal.

NONSPECIFIC AEROBIC TRAINING

While excessive nonspecific aerobic training is not a good time investment or a desired training stimulus, a moderate amount of generalized aerobic training is beneficial, especially for roped climbers. Whether your preference is running, cycling, or rowing, the

many systemic and local adaptations will increase both your aerobic power and capacity. The payoffs on the rock will be faster recovery at mid-climb rests and between boulder problems, improved recovery between redpoint attempts, and stamina to climb well throughout a long day at the sport crag or on multipitch gear routes.

The “right amount” of generalized aerobic training is a tough thing to specify, since your current level of conditioning, climbing goals, and time available all factor in. As a very rough guideline, however, I suggest doing two or three sessions of moderate-intensity aerobic activity per week, each lasting twenty to forty minutes. Certainly there's no reason, or benefit, for a boulderer or sport climber to perform excessive aerobic training such as running thirty-plus miles (fifty-plus km) per week—such a program will likely hurt your climbing-specific strength and power and may leave you too fatigued to climb your best.

Another aerobic training option is sprint intervals, performed on a rowing machine or on foot (running). Alternating between twenty seconds of near-all-out sprinting (90 percent intensity) and forty seconds of easy “recovery” jogging is an excellent stimulus for gains in aerobic power and capacity. This is a rigorous way to go about your aerobic training, and it should be limited to twice per week.

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